Data Bank — SQL Case Study #4

8 Week SQL Challenge by Danny Ma

Section A: Customer Nodes Exploration

A1. How many unique nodes are there on the Data Bank system?

SQL Query:

```
SELECT COUNT(DISTINCT node_id) AS nodes
FROM customer_nodes;
```

Output:

	nodes bigint
1	5

A2. What is the number of nodes per region?

SQL Query:

```
SELECT region_id, COUNT(node_id) AS nodes
FROM customer_nodes
GROUP BY region_id;
```

Output:

	region_id integer	nodes bigint
1	1	770
2	3	714
3	5	616
4	4	665
5	2	735

A3. How many customers are allocated to each region?

SQL Query:

```
SELECT region_id, COUNT(DISTINCT customer_id) AS nodes
FROM customer_nodes
GROUP BY region_id;
```

Output:

	region_id integer	nodes bigint
1	1	110
2	2	105
3	3	102
4	4	95
5	5	88

A4. How many days on average are customers reallocated to a different node?

SQL Query:

```
WITH cte AS (
    SELECT customer_id,
           start_date,
           end_date,
           LEAD(start_date) OVER (
               PARTITION BY customer_id
               ORDER BY start_date
           ) AS next_start_date
    FROM data_bank.customer_nodes
)
SELECT ROUND(AVG(end_date - start_date)) AS avg_days_reallocated
FROM cte
WHERE next_start_date IS NOT NULL
  AND end_date < next_start_date;
```

Output:

	avg_days_reallocated numeric
1	15

A5. What is the median, 80th and 95th percentile for this same reallocation days metric for each region?

SQL Query:

```
WITH reallocation AS (
    SELECT region_id,
           end_date - start_date AS reallocation_days
    FROM (
        SELECT region_id,
               start_date,
               end_date,
               LEAD(start_date) OVER (
                   PARTITION BY customer_id
                   ORDER BY start_date
               ) AS next_start_date
        FROM data_bank.customer_nodes
    ) t
    WHERE next_start_date IS NOT NULL
      AND end_date < next_start_date
)
SELECT
    region_id,
    PERCENTILE_CONT(0.5)  WITHIN GROUP (ORDER BY reallocation_days) AS median,
    PERCENTILE_CONT(0.8)  WITHIN GROUP (ORDER BY reallocation_days) AS p80,
    PERCENTILE_CONT(0.95) WITHIN GROUP (ORDER BY reallocation_days) AS p95
FROM reallocation
GROUP BY region_id
ORDER BY region_id;
```

Output:

	region_id integer	median double precision	p80 double precision	p95 double precision
1	1	15	23	28
2	2	15	23	28
3	3	15	24	28
4	4	15	23	28
5	5	15	24	28

Section B: Customer Transactions

B1. What is the unique count and total amount for each transaction type?

SQL Query:

```
SELECT
    txn_type,
    COUNT(DISTINCT customer_id) AS times_done,
    SUM(txn_amount) AS total_amount
FROM customer_transactions
GROUP BY txn_type;
```

Output:

	txn_type character varying (10)	times_done bigint	total_amount bigint
1	deposit	500	1359168
2	purchase	448	806537
3	withdrawal	439	793003

B2. What is the average total historical deposit counts and amounts for all customers?

SQL Query:

```
SELECT
    ROUND(COUNT(*)::numeric / COUNT(DISTINCT customer_id), 0) AS avg_deposit_count,
    ROUND(SUM(txn_amount) / COUNT(DISTINCT customer_id), 2) AS avg_deposit_amount
FROM customer_transactions
WHERE txn_type = 'deposit';
```

Output:

	avg_deposit_count numeric	avg_deposit_amount numeric
1	5	2718.34

B3. For each month — how many Data Bank customers make more than 1 deposit and either 1 purchase or 1 withdrawal in a single month?

SQL Query:

```
WITH monthly_hst AS (
    SELECT
        customer_id,
        EXTRACT(YEAR FROM txn_date) AS txn_year,
        EXTRACT(MONTH FROM txn_date) AS txn_month,
```

```
        COUNT(CASE WHEN txn_type = 'deposit' THEN 1 END) AS deposit_count,
        COUNT(CASE WHEN txn_type = 'purchase' THEN 1 END) AS purchase_count,
        COUNT(CASE WHEN txn_type = 'withdrawal' THEN 1 END) AS withdrawal_count
    FROM customer_transactions
    GROUP BY customer_id, EXTRACT(YEAR FROM txn_date), EXTRACT(MONTH FROM txn_date)
)
SELECT
    txn_year,
    txn_month,
    COUNT(DISTINCT customer_id) AS total_customers
FROM monthly_hst
WHERE deposit_count > 1
  AND (purchase_count = 1 OR withdrawal_count = 1)
GROUP BY txn_year, txn_month
ORDER BY txn_year, txn_month;
```

Output:

	txn_year numeric	txn_month numeric	total_customers bigint
1	2020	1	115
2	2020	2	108
3	2020	3	113
4	2020	4	50

B4. What is the closing balance for each customer at the end of the month?

SQL Query:

```
WITH monthly_hst AS (
    SELECT
        customer_id,
        EXTRACT(YEAR FROM txn_date) AS txn_year,
        EXTRACT(MONTH FROM txn_date) AS txn_month,
        SUM(CASE WHEN txn_type = 'deposit' THEN txn_amount ELSE 0 END) AS deposit_total,
        SUM(CASE WHEN txn_type = 'purchase' THEN txn_amount ELSE 0 END) AS purchase_total,
        SUM(CASE WHEN txn_type = 'withdrawal' THEN txn_amount ELSE 0 END) AS
withdrawal_total
    FROM customer_transactions
    GROUP BY customer_id, EXTRACT(YEAR FROM txn_date), EXTRACT(MONTH FROM txn_date)
)
SELECT
    customer_id,
    txn_year,
    txn_month,
    deposit_total - (purchase_total + withdrawal_total) AS closing_amount
FROM monthly_hst
ORDER BY customer_id;
```

Output:

	customer_id integer	txn_year numeric	txn_month numeric	closing_amount bigint
1	1	2020	1	312
2	1	2020	3	-952
3	2	2020	3	61
4	2	2020	1	549
5	3	2020	1	144
6	3	2020	3	-401
7	3	2020	2	-965
8	3	2020	4	493
9	4	2020	1	848
10	4	2020	3	-193
11	5	2020	1	954
12	5	2020	4	-490
13	5	2020	3	-2877
14	6	2020	3	392
15	6	2020	2	-785

B5. What is the percentage of customers who increase their closing balance by more than 5%?

SQL Query:

```
WITH monthly_hst AS (
    SELECT
        customer_id,
        EXTRACT(YEAR FROM txn_date) AS txn_year,
        EXTRACT(MONTH FROM txn_date) AS txn_month,
        SUM(CASE WHEN txn_type = 'deposit' THEN txn_amount ELSE 0 END) AS deposit_total,
        SUM(CASE WHEN txn_type = 'purchase' THEN txn_amount ELSE 0 END) AS purchase_total,
        SUM(CASE WHEN txn_type = 'withdrawal' THEN txn_amount ELSE 0 END) AS
withdrawal_total
    FROM customer_transactions
    GROUP BY customer_id, EXTRACT(YEAR FROM txn_date), EXTRACT(MONTH FROM txn_date)
),
monthly_balance AS (
    SELECT
        customer_id,
        txn_year,
        txn_month,
        deposit_total - (purchase_total + withdrawal_total) AS closing_amount,
        LAG(deposit_total - (purchase_total + withdrawal_total))
            OVER (PARTITION BY customer_id ORDER BY txn_year, txn_month) AS
prev_closing_amount
    FROM monthly_hst
)
SELECT
    ROUND(
        COUNT(customer_id) * 100.0
```

```
        / (SELECT COUNT(DISTINCT customer_id) FROM customer_transactions),
        2
    ) AS increase_by5
FROM monthly_balance
WHERE prev_closing_amount IS NOT NULL
  AND (closing_amount - prev_closing_amount) / prev_closing_amount > 0.05;
```

Output:

	increase_by5 numeric
1	37.20